

2024 PH1012 Answers for Mid-term 1

1. 225 days, many got the wrong answer 36 days.
2. E
3. Solve the problem in 2 stages:
 Stage 1: $v_1^2 = u^2 + 2ay$
 Stage 2: $v_F^2 = v_1^2 - 2gy_m$ where $v_F = 0$ at the greatest height y_m

0.75 km

4. $v_f^2 = (v \cos \theta)^2 + [(v \sin \theta)^2 - 2gh]$
 $v_f = 5.0 \text{ m/s}$
5.
 $T_1 \cos 30^\circ + T_2 \cos 60^\circ - mg = 0$, $T_1 \sin 30^\circ + T_2 \sin 60^\circ = 0$
 $T_2 = 24.5 \text{ N}$
6. E
7. $9.6 \times 10^{-4} \text{ m}$
8. $W_{car} + W_{ice} - U = 0$
 $A = 24 \text{ m}^2$
9. C

Long Questions

Q1.a i) $pV = nRT$ 0.468 g; ii) 61°C

$$\text{b) } m_{ice} (c_{ice}(5) + l_f + c_{water}(21)) = m_{drink} c_{water}(5)$$

$m_{ice} = 14.6 \text{ g}$, 12 pieces

$$\text{Q2. a) } t_{rock} = 1.87 \text{ s}; \quad t_{boat} = 1.67 \text{ s}$$

$$x_{rock} = 2 \cos 50^\circ \times 1.87 = 2.87 \text{ m}$$

$2.87 \text{ m} < \frac{10-3}{2} = 3.5 \text{ m}$ so the rock will not hit the boat.

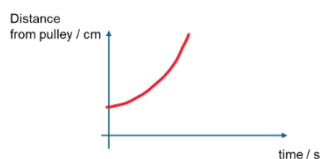
b) A. Increasing to more than 50° below the horizontal.

Q3 a)

$$Mg - T = Ma - (1)$$

$$T + 2Mg \sin 30^\circ - \mu 2Mg \cos 30^\circ = 2Ma - (2)$$

$$A = 5.40 \text{ m/s}^2$$



- b)
- c) Larger angle, steeper; smoother slope; replace M (vertical) with larger mass